Government of Karnataka
Department of Collegiate education

GOVERNMENT FIRST GRADE COLLEGE, NAVANAGAR, BAGALKOT-587103

Department of Computer Science

BCA IV Semester (SEP)

Web Technology LAB
2025-26

Formative Assessment: 10 Marks **S**ummative Assessment: 40 Marks

INDEX				
SL. NO	**PROGRAMS**	**DATE**	**PAGE NO.**	**SIGN**
	PART-A			
1.	Write HTML – Create a personal portfolio webpage with a bio, profile image, and social media links			
2.	HTML – Design a student registration form with input fields for name, email, Write gender, and course selection			
3.	Write HTML – Develop a timetable using tables and include rowspan and colspan for better structuring.			
4.	Write HTML and CSS – Build a responsive navigation bar using Flexbox with hover effects.			
5.	Write HTML and CSS – Design a modern login form with styled input fields, buttons, and shadows.			
6.	Write HTML and CSS – Create a responsive card layout using CSS Grid and Flexbox for a product showcase.			
7.	Write JavaScript – Implement a simple interest calculator that takes input and displays the result dynamically			
8.	Write JavaScript – Create a to-do list application where users can add, remove, and mark tasks as completed.			
9.	Write JavaScript – Filter and search a list ofstudents dynamically using filter() and map().			
10.	Write JavaScript – Build a temperature converter that switches between Celsius and Fahrenheit.			

PART-B

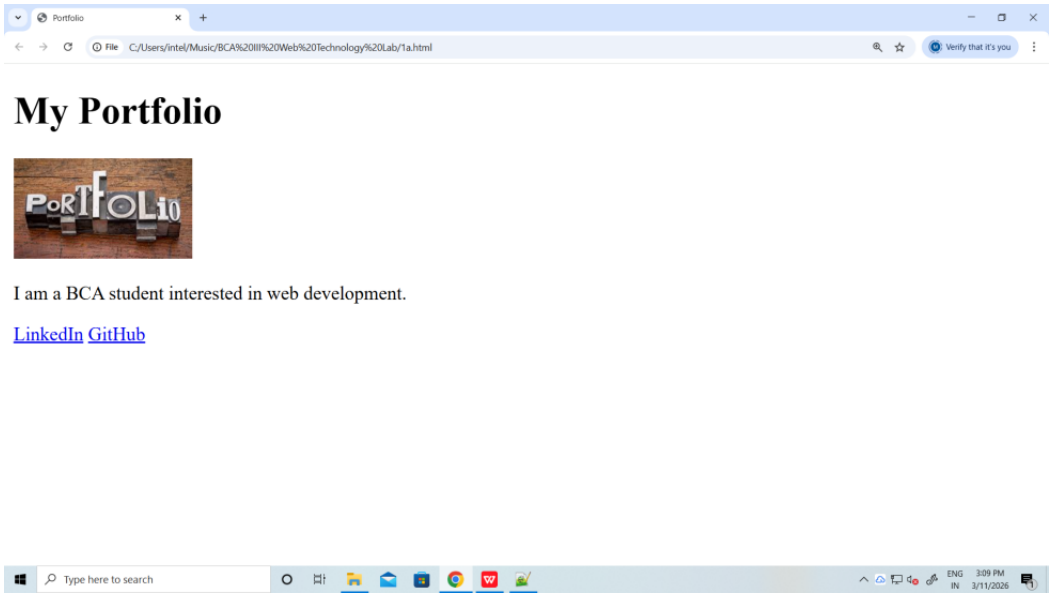
11.	Write JavaScript – Implement a shopping cart where adding/removing items updates the total price dynamically			
12.	Write JavaScript – Change background color dynamically on a button click using JavaScript.			
13.	Write JavaScript – Validate a registration form for correct username format, password strength, and empty fields.			
14.	Write JavaScript – Display a real-time digital clock using setInterval().			
15.	Write JavaScript – Implement a countdown timer that stops at zero using setTimeout().			

PART - A

1. **Write HTML – Create a personal portfolio webpage with a bio, profile image, and social media links**

```
<!DOCTYPE html>
<html>
<head><title>Portfolio</title></head>
<body>
<h1>My Portfolio</h1>
<img src="C:\Users\intel\Music\BCA III Web Technology Lab\1.jfif" width="150">
<p>I am a BCA student interested in web development.</p>
<a href="https://in.linkedin.com">LinkedIn</a>
<a href="https://github.com">GitHub</a>
</body>
</html>
```

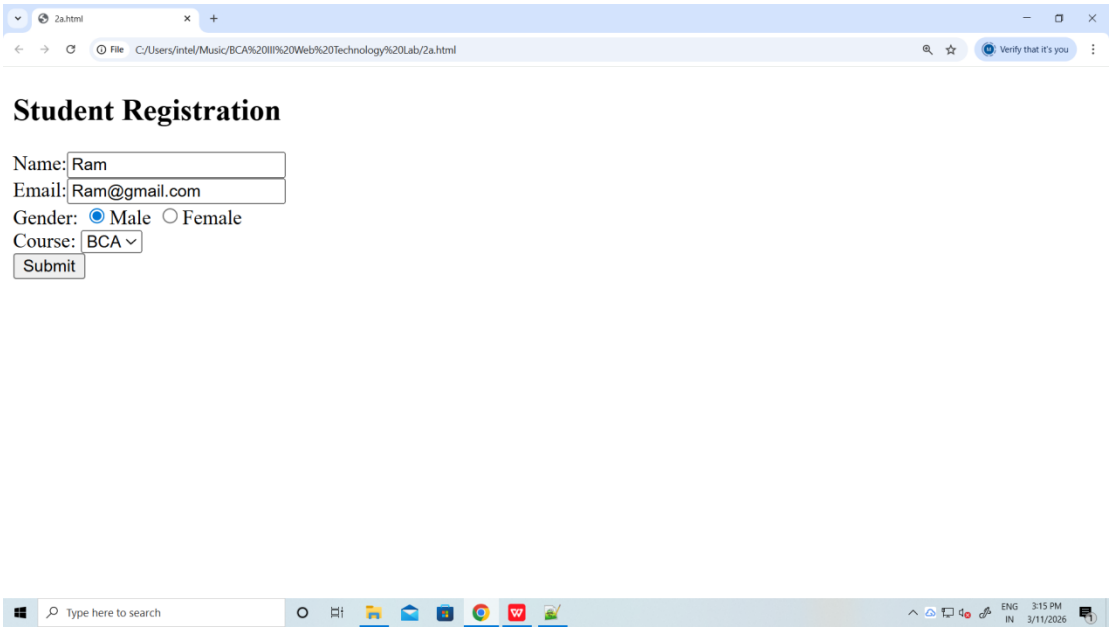
Output :


Portfolio
File C:/Users/intel/Music/BCA%20III%20Web%20Technology%20Lab/1a.html
Verify that it's you
My Portfolio
I am a BCA student interested in web development.
LinkedIn GitHub
Type here to search
ENG IN
3:09 PM 3/11/2026

2. **HTML – Design a student registration form with input fields for name, email, Write gender, and course selection**

```
<!DOCTYPE html>
<html>
<body>
<h2>Student Registration</h2>
<form>
Name:<input type="text"><br>
Email:<input type="email"><br>
Gender:
<input type="radio" name="g">Male
<input type="radio" name="g">Female<br>
Course:
<select>
<option>BCA</option>
<option>BBA</option>
<option>BSc</option>
</select><br>
<input type="submit">
</form>
</body>
</html>
```

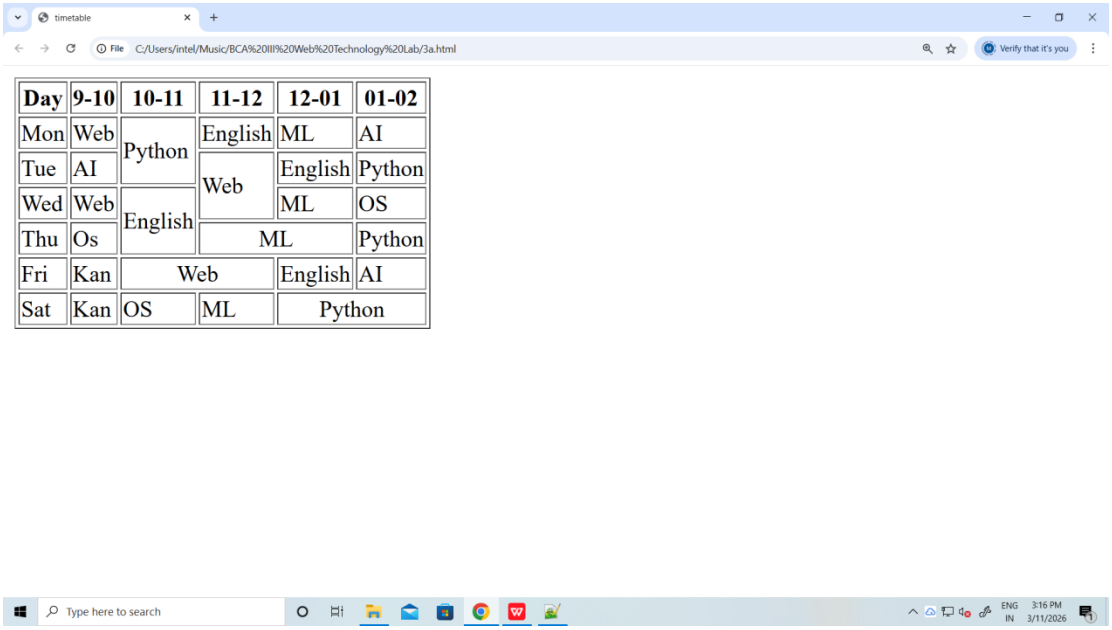
Output:


2a.html
File C:/Users/intel/Music/BCA%20III%20Web%20Technology%20Lab/2a.html
Verify that it's you
Student Registration
Name: Ram
Email: Ram@gmail.com
Gender: Male Female
Course: BCA
Submit
Type here to search
ENG IN
3:15 PM 3/11/2026

3. **Write HTML – Develop a timetable using tables and include rowspan and colspan for better structuring.**

```
<!DOCTYPE html>
<html>
<head><title>timetable</title></head>
<body>
<table border="1">
<tr><th>Day</th><th>9-10</th><th>10-11</th><th>11-12</th><th>12-01</th><th>01-02</th></tr>
<tr><td>Mon</td><td>Web</td><td
rowspan="2">Python</td><td>English</td><td>ML</td><td>AI</td></tr>
<tr><td>Tue</td><td>AI</td><td rowspan="2">Web</td><td>English</td><td>Python</td></tr>
<tr><td>Wed</td><td>Web</td><td rowspan="2">English</td><td>ML</td><td>OS</td></tr>
<tr><td>Thu</td><td>Os</td><td colspan="2"><center>ML</td></center><td>Python</td></tr></tr>
<tr><td>Fri</td><td>Kan</td><td
colspan="2"><center>Web</td></center><td>English</td><td>AI</td></tr></tr>
<tr><td>Sat</td><td>Kan</td><td>OS</td><td>ML</td><td
colspan="2"><center>Python</td></center></tr></tr>
</table>
</body>
</html>
```

Output:



Day	9-10	10-11	11-12	12-01	01-02
Mon	Web	Python	English	ML	AI
Tue	AI		Web	English	Python
Wed	Web	English		ML	OS
Thu	Os		ML		Python
Fri	Kan	Web		English	AI
Sat	Kan	OS	ML	Python	

4. **Write HTML and CSS – Build a responsive navigation bar using Flexbox with hover effects.**

```
<!DOCTYPE html>
<html>
<head><title>Navigation</title></head>
<body>
<style>
nav{display:flex;background:#333}
nav a{color:white;padding:10px;text-decoration:none}
nav a:hover{background:orange}
</style>
<nav>
<a href="index.html">Home</a>
<a href="about.html">About</a>
<a href="contact.html">Contact</a>
</nav>
<p>Success is not the key to happiness. Happiness is the key to success.</p>
<img src="C:\Users\intel\Music\BCA III Web Technology Lab\11.jfif" width="550">
</body>
</html>
```

Index.html

```
<!DOCTYPE html>
<html>
<head><title>Index</title></head>
<body>
<h1>Welcome to My Website</h1>
<img src="C:\Users\intel\Music\BCA III Web Technology Lab\3.png" width="550"><br>
<button onclick="window.history.back()">Go Back</button>
</body>
</html>
```

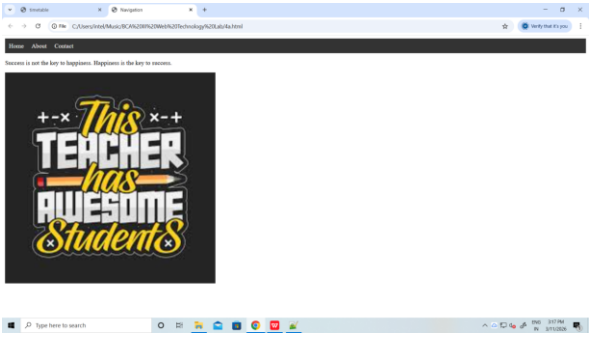
About.html

```
<!DOCTYPE html>
<html>
<head><title>About Us</title></head>
<section>
        <h2>About Us</h2>
        <p>This is a paragraph about the content on the home page. We use HTML to structure our content.</p>
                        <img src="C:\Users\intel\Music\BCA III Web Technology Lab\2.webp" width="550"><br>
                        <button onclick="window.history.back()">Go Back</button>
    </section>
</body>
</html>
```

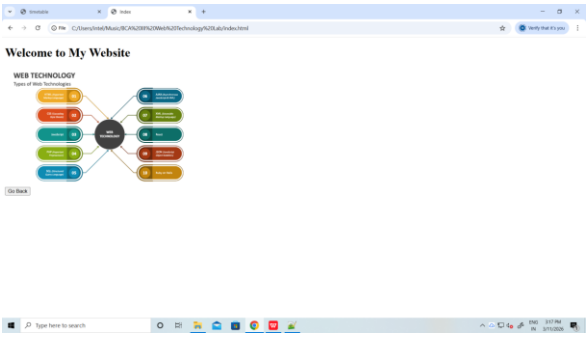
Contact.html

```
<!DOCTYPE html>
<html>
<head><title>Contact Us</title></head>
<body>
<p> Phone Number : 0123456789<p/>
<p >Email Id : abx@gmail.com <p/>
<img src="C:\Users\intel\Music\BCA III Web Technology Lab\01.jpg" width="660"><br>
<button onclick="window.history.back()">Go Back</button>
</body>
</html>
```

Output:

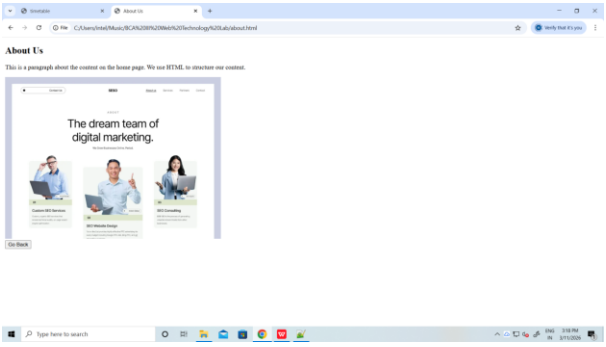


Output:



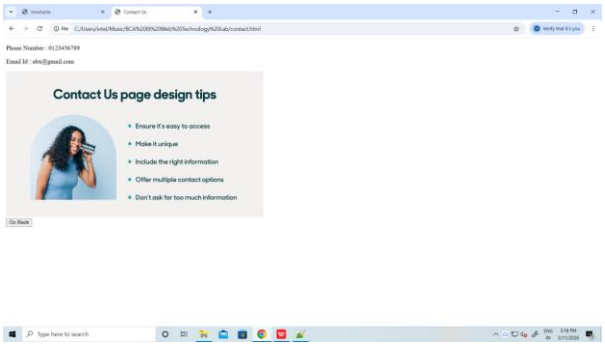
Home Page

Output:



About Page

Output:

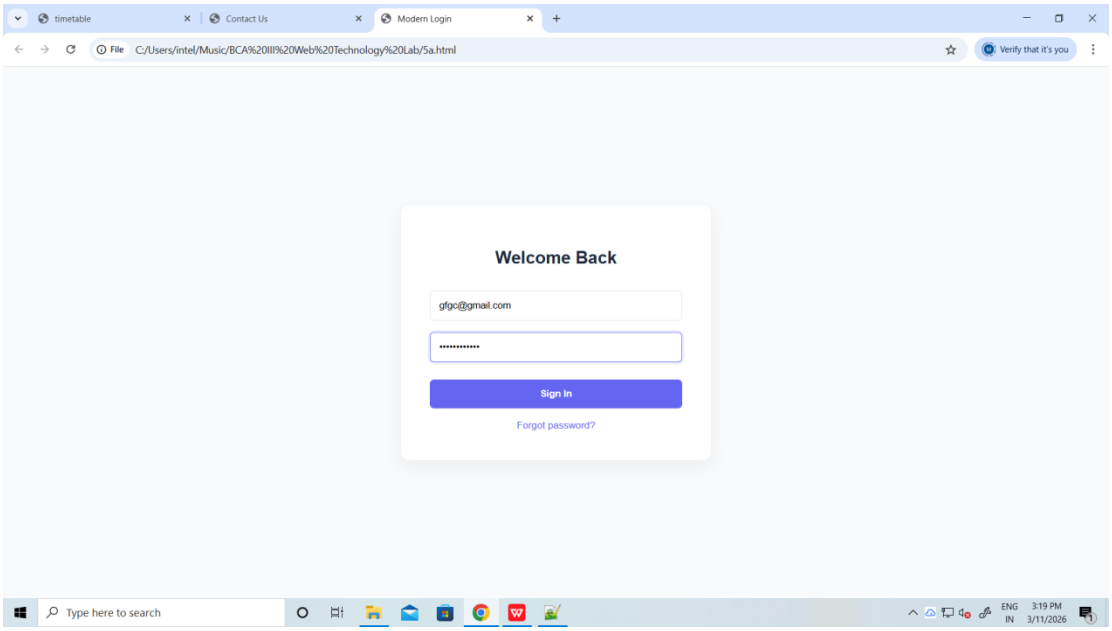


Contact Us Page

5. **Write HTML and CSS – Design a modern login form with styled input fields, buttons, and shadows.**

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Modern Login</title>
   <style>
      :root { --p: #6366f1;
                  --bg: #f8fafc;
                  }
      body { font-family: sans-serif; background: var(--bg);
                  display: flex; align-items: center;
                  justify-content: center; height: 100vh; margin: 0;
                  }
      .card { background: #fff; padding: 2.5rem; border-radius: 12px; box-shadow: 0 10px 25px
rgba(0,0,0,0.05); width: 100%;
                  max-width: 350px; text-align: center; }
      h2 { margin-bottom: 1.5rem; color: #1e293b; }
      input { width: 100%; padding: 12px;
                  margin: 8px 0;
                  border: 1px solid #e2e8f0;
                  border-radius: 6px;
                  box-sizing: border-box;
                  outline: none; }
      input:focus { border-color: var(--p);
                  box-shadow: 0 0 0 3px rgba(99, 102, 241, 0.1); }
      button { width: 100%; background: var(--p);
                  color: #fff; padding: 12px; border: none;
                  border-radius: 6px; cursor: pointer; font-weight: 600; margin-top: 1rem; transition: 0.2s;
}
      button:hover { opacity: 0.9; }
      a { font-size: 0.85rem; color: var(--p); text-decoration: none; display: block; margin-top: 1rem; }
   </style>
</head>
<body>
   <div class="card">
      <h2>Welcome Back</h2>
      <form action="#" method="POST">
         <input type="email" placeholder="Email Address" required>
         <input type="password" placeholder="Password" required>
         <button type="submit">Sign In</button>
      </form>
      <a href="#">Forgot password?</a>
   </div>
</body> </html>
```

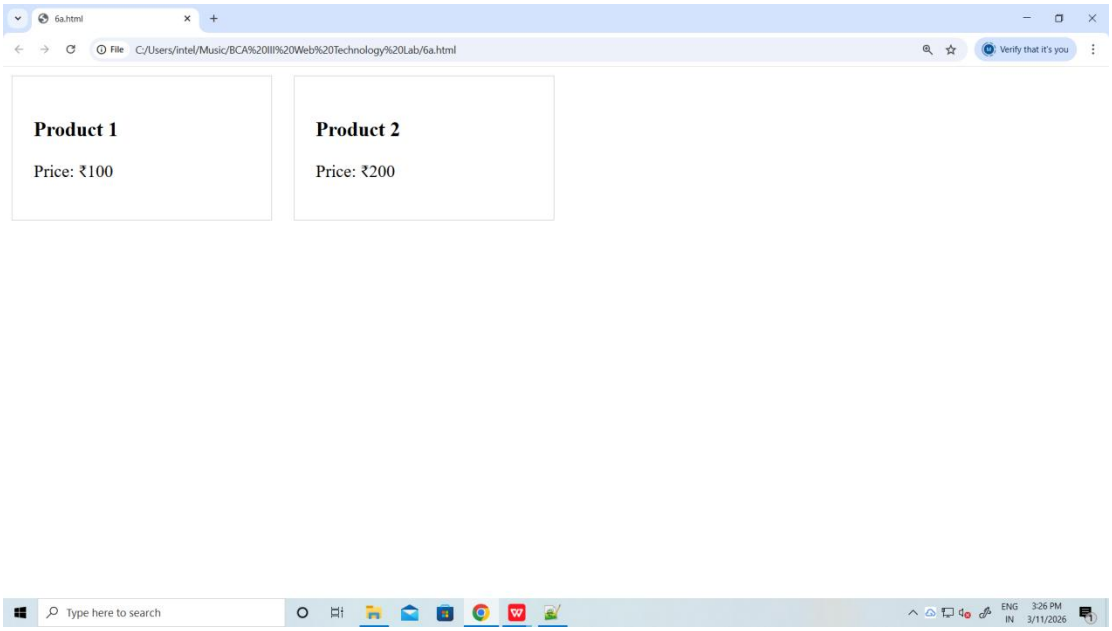
Output:


timetable
Contact Us
Modern Login
File
C:/Users/intel/Music/BCA%20III%20Web%20Technology%20Lab/5a.html
Verify that it's you
Welcome Back
gfgc@gmail.com
Sign In
Forgot password?
Type here to search
ENG
IN
3:19 PM
3/11/2026

6. **Write HTML and CSS – Create a responsive card layout using CSS Grid and Flexbox for a product showcase.**

```html
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <link rel="stylesheet" href="styles.css">
   <title>Responsive Card Layout</title>
         <style>

</head>
<body>
   <div class="card-container">
      <div class="card">
         <img src="image1.jpg" alt="Card 1">
         <h2>Card 1</h2>
         <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit.</p>
      </div>
      <div class="card">
         <img src="image2.jpg" alt="Card 1">
         <h2>Card 2</h2>
         <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit.</p>
      </div>
      <div class="card">
         <img src="image3.jpg" alt="Card 1">
         <h2>Card 3</h2>
         <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit.</p>
      </div>
      <!-- Repeat the card structure for more items -->
   </div>
</body>
</html>
```

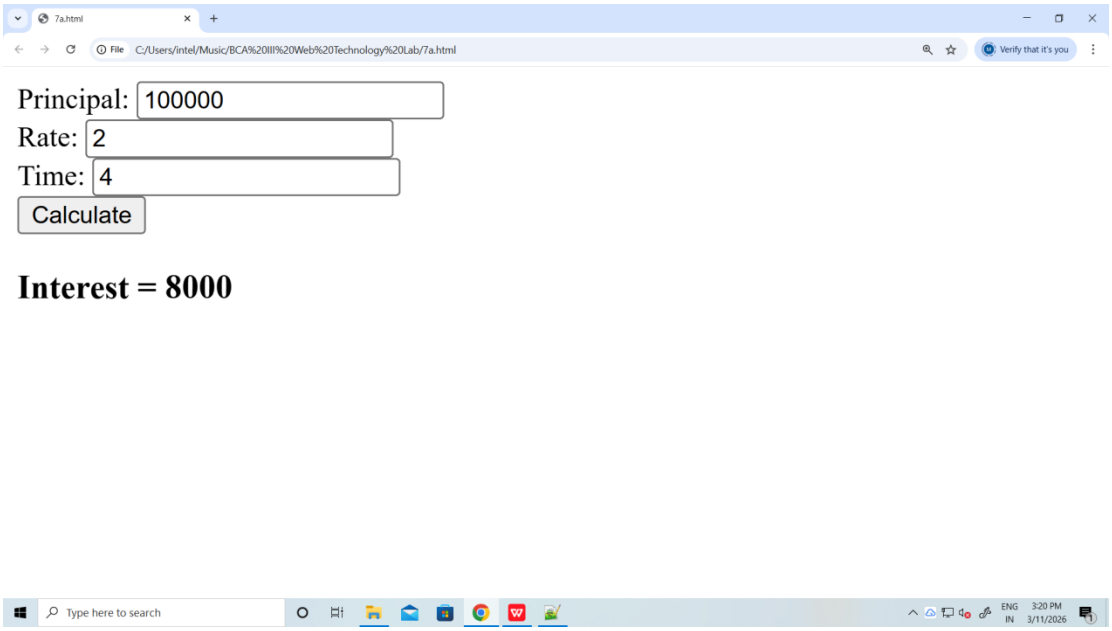
Output:


6a.html
File C:/Users/intel/Music/BCA%20III%20Web%20Technology%20Lab/6a.html
Verify that it's you
Product 1
Price: ₹100
Product 2
Price: ₹200
Type here to search
ENG IN
3:26 PM 3/11/2026

7. **Write JavaScript – Implement a simple interest calculator that takes input and displays the result dynamically**

```
<!DOCTYPE html>
<html>
<body>

Principal: <input id="p"><br>
Rate: <input id="r"><br>
Time: <input id="t"><br>

<button onclick="calc()">Calculate</button>

<h3 id="result"></h3>
<script>
function calc(){
let p=document.getElementById("p").value;
let r=document.getElementById("r").value;
let t=document.getElementById("t").value;

let si=(p*r*t)/100;
document.getElementById("result").innerHTML="Interest = "+si;
}
</script>
</body>
</html>
```

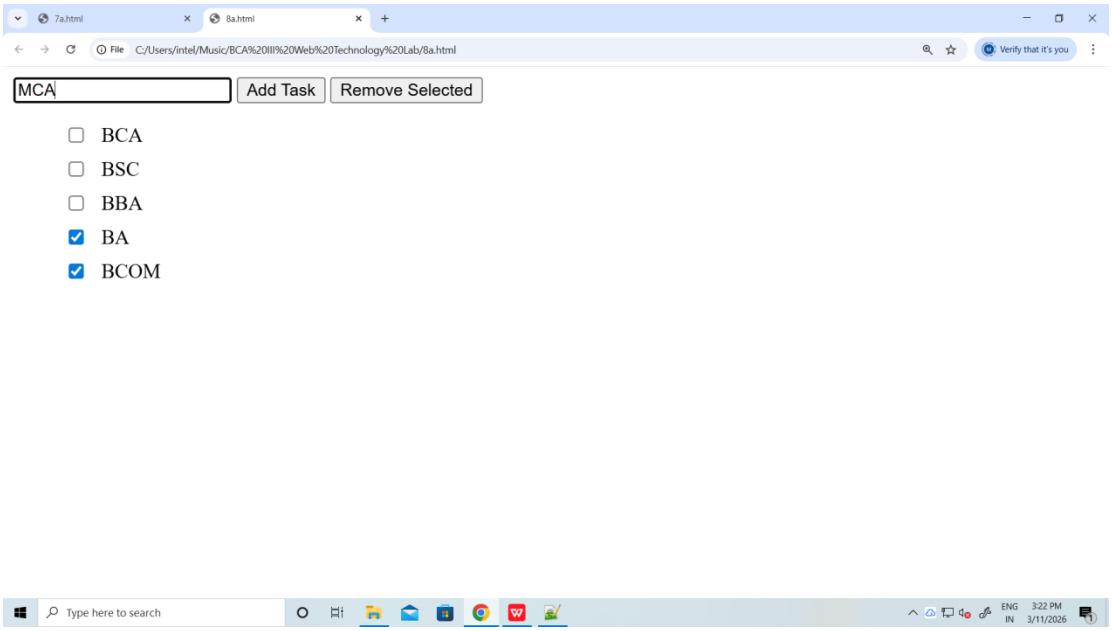
Output:


7a.html
File C:/Users/intel/Music/BCA%20III%20Web%20Technology%20Lab/7a.html
Verify that it's you
Principal: 100000
Rate: 2
Time: 4
Calculate
Interest = 8000
Type here to search
ENG IN
3:20 PM 3/11/2026

8. **Write JavaScript – Create a to-do list application where users can add, remove, and mark tasks as completed.**

```
<!DOCTYPE html>
<html>
<head>
<style>
  .completed { text-decoration: line-through; color: gray; }
  li { margin-bottom: 8px; list-style: none; display: flex; align-items: center; gap: 10px; }
</style>
</head>
<body>

  <input id="task" placeholder="Enter a task...">
  <button onclick="addTask()">Add Task</button>
  <button onclick="removeSelected()">Remove Selected</button>
  <ul id="list"></ul>
<script>
function addTask() {
  let taskInput = document.getElementById("task");
  let taskValue = taskInput.value;

  if (taskValue.trim() === "") return;

  let li = document.createElement("li");

  li.innerHTML = `
    <input type="checkbox" class="task-check">
    <span onclick="markTask(event)">${taskValue}</span>
  `;

  document.getElementById("list").appendChild(li);
  taskInput.value = "";
}
function removeSelected() {
  const checkboxes = document.querySelectorAll('.task-check:checked');

  checkboxes.forEach(checkbox => {
    checkbox.closest('li').remove();
  });
}
function markTask(event) {
  event.target.classList.toggle("completed");
}
</script>
</body>
</html>
```

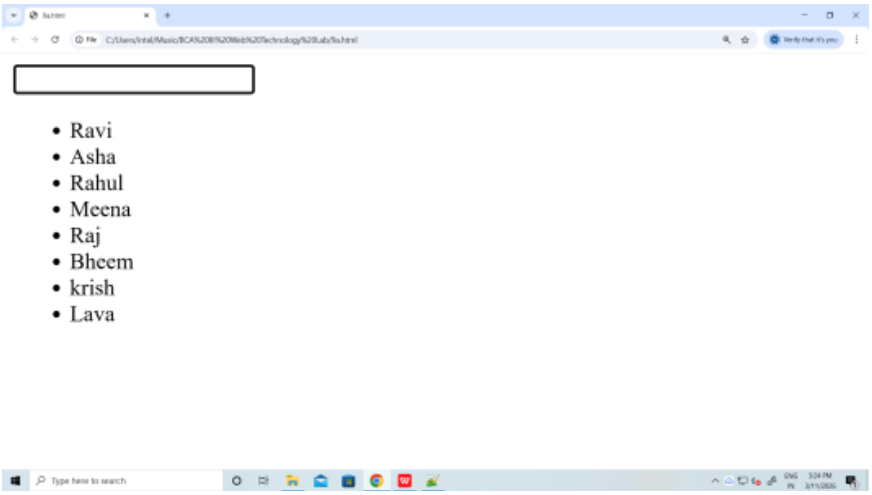
Output:

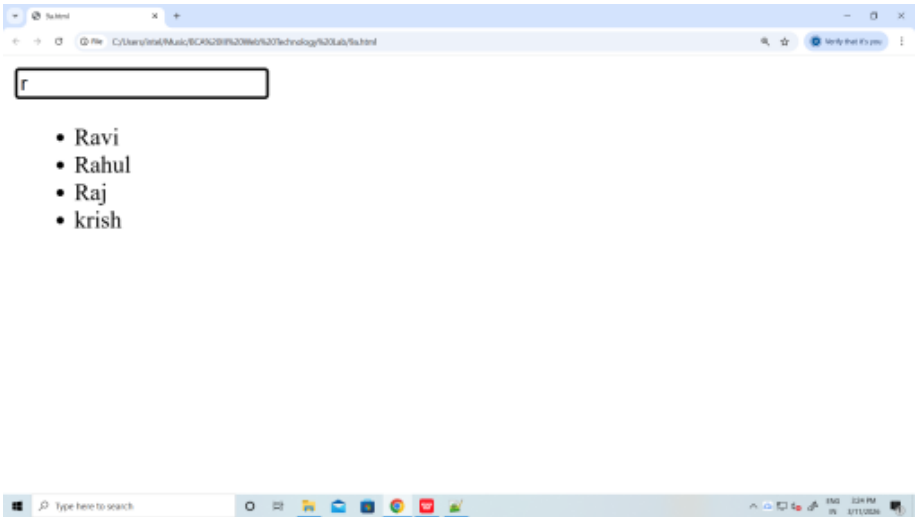

7a.html
8a.html
File
C:/Users/intel/Music/BCA%20III%20Web%20Technology%20Lab/8a.html
Verify that it's you
MCA
Add Task
Remove Selected
BCA
BSC
BBA
BA
BCOM
Type here to search
ENG
IN
3:22 PM
3/11/2026

9. Write JavaScript – Filter and search a list ofstudents dynamically using filter() and map().

```
<!DOCTYPE html>
<html>
<body>

<input id="search" onkeyup="filter()">

<ul id="students">
<li>Ravi</li>
<li>Asha</li>
<li>Rahul</li>
<li>Meena</li>
<li>Raj</li>
<li>Bheem</li>
<li>krish</li>
<li>Lava</li>
</ul>

<script>
function filter(){
let input=document.getElementById("search").value.toLowerCase();
let items=document.querySelectorAll("#students li");

items.forEach(i=>{
i.style.display=i.innerText.toLowerCase().includes(input) ? "" : "none";
});
}
</script>

</body>
</html>
```

Output:


Ravi
Asha
Rahul
Meena
Raj
Bheem
krish
Lava


r
Ravi
Rahul
Raj
krish

10. Write JavaScript – Build a temperature converter that switches between Celsius and Fahrenheit.

```
<!DOCTYPE html>
<html>
<body>

  <h2>Temperature Converter</h2>

  <input id="c" placeholder="Celsius">
  <button onclick="toFahrenheit()">Convert to °F</button>
  <p id="resultF"></p>

  <hr>

  <input id="fInput" placeholder="Fahrenheit">
  <button onclick="toCelsius()">Convert to °C</button>
  <p id="resultC"></p>

  <script>
  function toFahrenheit() {
    let c = Number(document.getElementById("c").value);
    let f = (c * 9/5) + 32;
    document.getElementById("resultF").innerHTML = f.toFixed(1) + " °F";
  }

  function toCelsius() {
    let f = Number(document.getElementById("fInput").value);
    let c = (f - 32) * 5/9;
    document.getElementById("resultC").innerHTML = c.toFixed(1) + " °C";
  }
  </script>

</body>
</html>
```

Output:

9a.html
10a.html
File
C:/Users/intel/Music/BCA%20III%20Web%20Technology%20Lab/10a.html
Verify that it's you
Temperature Converter
100
Convert to °F
212.0 °F
66
Convert to °C
18.9 °C
Type here to search
ENG
IN
3:25 PM
3/11/2026

Part - B

11. Write JavaScript – Implement a shopping cart where adding/removing items updates the total price dynamically

```
<!DOCTYPE html>
<html>
<head>
<title>Simple Shopping Cart</title>
<style>
table, th, td {
border:1px solid black;
border-collapse:collapse;
padding:8px;
text-align:center;
}
button{
padding:5px 10px;
}
</style>
</head>
<body>

<h2>Shopping Cart</h2>

<table>
<tr>
<th>Item</th>
<th>Price</th>
<th>Quantity</th>
<th>Action</th>
</tr>

<tr>
<td>Apple</td>
<td>₹50</td>
<td id="qty1">0</td>
<td>
<button onclick="addItem(1,50)">Add</button>
<button onclick="removeItem(1,50)">Remove</button>
</td>
</tr>

<tr>
<td>Banana</td>
<td>₹30</td>
<td id="qty2">0</td>
<td>
```

```
<button onclick="addItem(2,30)">Add</button>
<button onclick="removeItem(2,30)">Remove</button>
</td>
</tr>
<tr>
<td>Mango</td>
<td>₹70</td>
<td id="qty3">0</td>
<td>
<button onclick="addItem(3,70)">Add</button>
<button onclick="removeItem(3,70)">Remove</button>
</td>
</tr>
</table>
<h3>Total Price: ₹ <span id="total">0</span></h3>

<script>

let quantities = {1:0,2:0,3:0};
let prices = {1:50,2:30,3:70};

function addItem(id,price){
quantities[id]++;
document.getElementById("qty"+id).innerHTML = quantities[id];
updateTotal();
}

function removeItem(id,price){
if(quantities[id] > 0){
quantities[id]--;
document.getElementById("qty"+id).innerHTML = quantities[id];
}
updateTotal();
}

function updateTotal(){
let total = 0;

for(let item in quantities){
total += quantities[item] * prices[item];
}

document.getElementById("total").innerHTML = total;
}

</script>
</body>
</html>
```

Output:

Shopping Cart

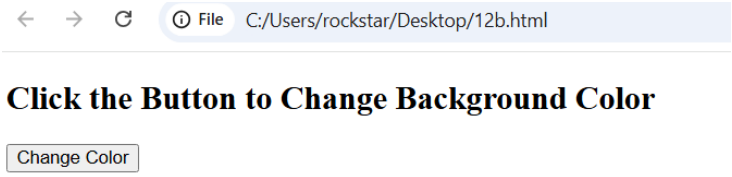
Item	Price	Quantity	Action
Apple	₹50	1	Add Remove
Banana	₹30	2	Add Remove
Mango	₹70	0	Add Remove

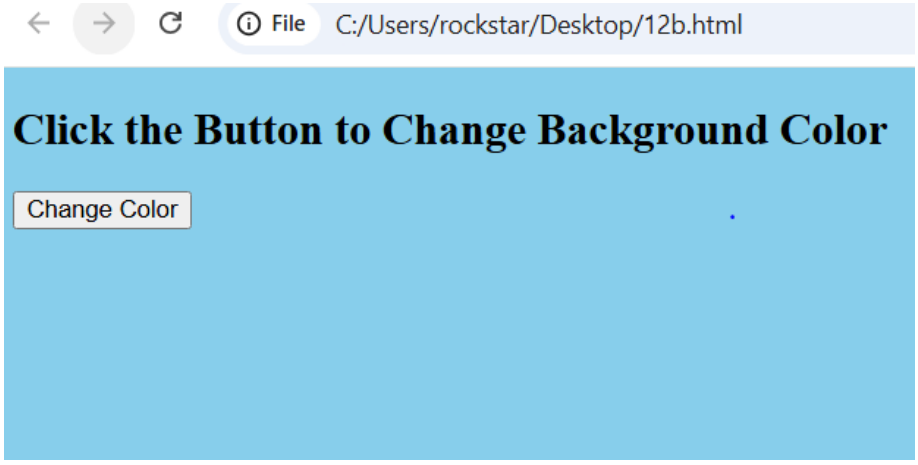
Total Price: ₹ 110

12. Write JavaScript – Change background color dynamically on a button click using JavaScript.

```
<!DOCTYPE html>
<html>
<head>
<title>Dynamic Background Color</title>
</head>
<body id="bg">
<h2>Click the Button to Change Background Color</h2>
<button onclick="changeColor()">Change Color</button>
<script>
let colors = [
"lightblue",
"lightgreen",
"yellow",
"orange",
"pink",
"violet",
"lightgray",
"skyblue",
"gold",
"coral"
];

function changeColor(){
let random = Math.floor(Math.random()*colors.length);
document.getElementById("bg").style.backgroundColor = colors[random];
}
</script>
</body>
</html>
```

Output:

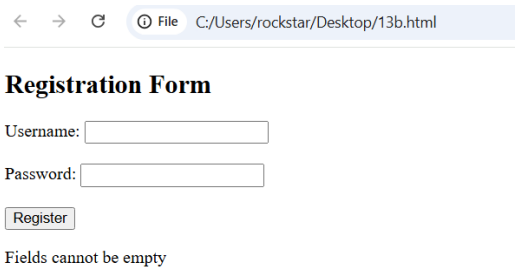

File
C:/Users/rockstar/Desktop/12b.html
Click the Button to Change Background Color
Change Color

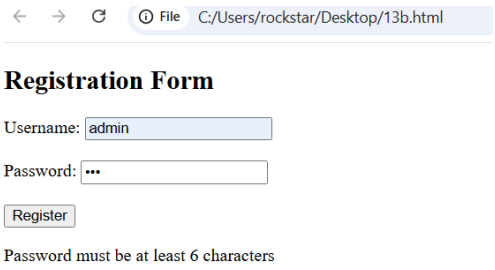

File
C:/Users/rockstar/Desktop/12b.html
Click the Button to Change Background Color
Change Color

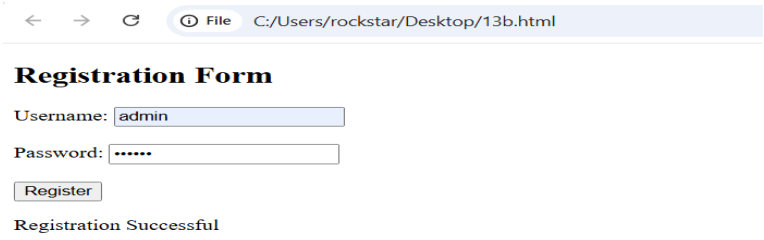
13. **Write JavaScript – Validate a registration form for correct username format, password strength, and empty fields.**

```
<!DOCTYPE html>
<html>
<head>
<title>Form Validation</title>
</head>
<body>

<h2>Registration Form</h2>

<form onsubmit="return validateForm()">

Username: <input type="text" id="username"><br><br>
Password: <input type="password" id="password"><br><br>

<input type="submit" value="Register">
</form>
<p id="msg"></p>
<script>
function validateForm(){
let user = document.getElementById("username").value;
let pass = document.getElementById("password").value;

if(user == "" || pass == ""){
document.getElementById("msg").innerHTML = "Fields cannot be empty";
return false;
}
if(pass.length < 6){
document.getElementById("msg").innerHTML = "Password must be at least 6 characters";
return false;
}
document.getElementById("msg").innerHTML = "Registration Successful";
return false;

}
</script>

</body>
</html>
```

Output:


File
C:/Users/rockstar/Desktop/13b.html
Registration Form
Username:
Password:
Register
Fields cannot be empty

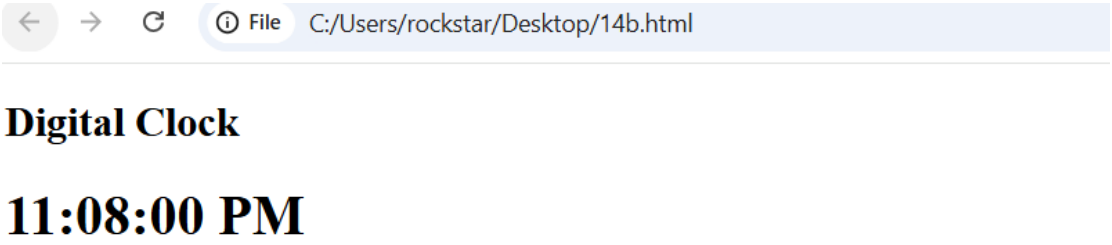

File
C:/Users/rockstar/Desktop/13b.html
Registration Form
Username: admin
Password: •••
Register
Password must be at least 6 characters


File
C:/Users/rockstar/Desktop/13b.html
Registration Form
Username: admin
Password: ••••••
Register
Registration Successful

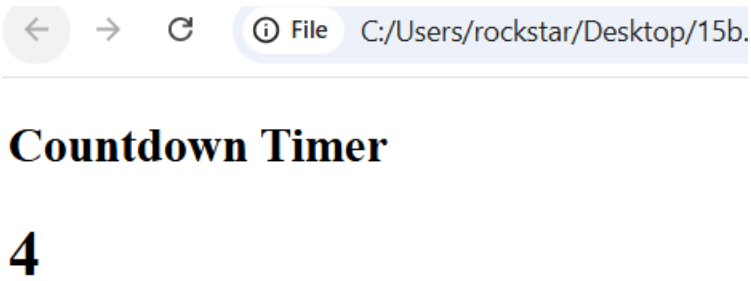
14. Write JavaScript – Display a real-time digital clock using setInterval().

```
<!DOCTYPE html>
<html>
<head>
<title>Digital Clock</title>
</head>
<body>
<h2>Digital Clock</h2>
<h1 id="clock"></h1>
<script>
setInterval(showTime,1000);
//1 seconds=1000 miliseconds
function showTime(){
let time = new Date();
document.getElementById("clock").innerHTML = time.toLocaleTimeString();
}

</script>
</body>
</html>
```

Output:


File
C:/Users/rockstar/Desktop/14b.html
Digital Clock
11:08:00 PM

15. Write JavaScript – Implement a countdown timer that stops at zero using setTimeout().

```
<!DOCTYPE html>
<html>
<head>
<title>Countdown Timer</title>
</head>
<body>
<h2>Countdown Timer</h2>
<h1 id="timer">10</h1>

<script>
let count = 10;
function countdown(){
document.getElementById("timer").innerHTML = count;
count--;

if(count >= 0){
setTimeout(countdown,1000);
}

}
countdown();
</script>
</body>
</html>
```

Output: